SW Engineering CSC648-848-05 Spring 2023
StudentSources

Team #04 “Fantastic Four”
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History Table:

M4v2T4	05/24/2023
M4v1T4	04/18/2023
M3v2T4	04/27/2023
M3v1T4	04/15/2023
M2v2T4	04/26/2023
M2v1T4	03/30/2023
M1v2T4	05/24/2023
M1v1T4	03/02/2023

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Data Definitions

General User: A general user is one who doesn't have permission to view any of the approved resources. As they also automatically lack an account on StudentSources, general users do not necessarily need to either log in nor do they have to register and have access to basic features. However, if they choose to do so, general users can sign up by providing an email address, password, and to make sure that they agree to the terms and conditions.

Registered User: Registered users are essentially users who have access to an account on StudentSources. Unlike general users, registered users have the capability to view resources that have been approved as well as have the capability to upload their own. Also, registered users are able to open their own file via creating separate tabs as long as they are logged in and registered into the system.

Account: An account is the one thing that differs between general users and registered users. Accounts store various forms of data belonging to those who are registered users, such as usernames, passwords, full names, and resources that said users have posted. Also, those who have accounts have full access to all approved and unapproved resources and have the capability to search for different files to be added to the list and/or share with other users.

Admin: Admin is a person/ team who is in charge of managing the website. The responsibilities of the admin include maintaining and troubleshooting the website, implementing security protocols, ensuring efficient user experience, and updating the website's contents. Admin also has authoritative power to approve or disapprove resources that users add to the website.

Resources: Resources are the contents added by the users in their account. The list of resources are text files, images/video, online contents. The maximum size of the resources a registered user shall upload would be 1 GB.

- **Text files:** Text files include files with extensions .pdf, and .txt.
- **Images:** Image content includes files with extension .jpg , .jpeg and .png.
- **Video:** Video content will be stored as a link.
- **Audio:** Audio content will be stored as a link.
- **Online Contents:** Online contents include the online resources such as link to webpage, YouTube videos.

Unapproved Resources: Unapproved resources are the resources that are yet to be approved by the admin or the resources that failed to acquire permission from the

admin to be added to the website. The criteria for resources to be approved include the relevancy of the content and if it is related to the course. The unapproved resources are listed as approved resources once they receive approval. Suppose the resources are unsuccessful in gaining permission. In that case, the resource is removed from the website, and the respective user of the resource is informed via the notification panel of the website.

Approved Resources: Approved resources are the resources that are approved based on the approval criteria. Once the resources are approved by the admin, the users can add the content to the list and share it with other users with the generated link.

Directory and Directory Tree: A directory is a folder that contains resources inside. A directory tree is the structure to organize the contents in the directory. Registered users can create multiple directories and subdirectories dedicated to different courses and topics. Registered users can use the default layout or create it for note organization. Registered users can label directories as professor's name, course name, or course abbreviation and number (CSC648, for example). Registered users can share the directories and subdirectories with other users with the generated link.

Comment: Comments are the remarks that users leave for uploaded resources. Only registered users shall be able to leave a comment on a resource. Registered users can leave one or multiple comments on their own resources, shared resources or publicly accessible resources. The posted comments are verified if appropriate by the admin. If comments are not appropriate, the admin can delete the comment from the resource.

Friend: Friend(s) are other registered users who are added as ‘Friend’ by a registered user. Adding other registered users as friends helps find users faster by applying filters while searching. A registered user has options to add/remove friends, and to list friends as ‘Favorite.’

Note: Note(s) are the record of brief descriptions and/or thoughts registered users can add to their accounts. A panel is available for registered users to add notes to their accounts. The associated note panel contains features where a registered user can type and save notes on the website.

Functional Requirements

Priority 1

General User

1.1. A general user shall be able to create one or more accounts.
1.2. A general user shall be able to sign up with a unique email address.
1.3. A general user shall be able to sign up with a non-unique password.
1.4. A general user shall be able to see if all requirements for creating a password are met.
1.5. A general user shall be able to sign up with a non-unique username.
1.6. A general user shall be able to read the terms and conditions.
1.7. A general user shall be able to agree to the terms and conditions.

Registered User

2.1. A registered user shall have at least one account.
2.3. A registered user shall be able to view their account information in the Account page.
2.4. A registered user shall be able to temporarily unhide account password information in the *Account* page. By default, when a registered user visits the *Account* page, the password would be hidden for security and password.
2.5. A registered user shall be able to change their account username.
2.6. A registered user shall be able to change their account email.
2.7. A registered user shall be able to change their account password.
2.8. A registered user shall be able to upload any supported file type to their account's cloud storage.
2.10. A registered user shall be able to navigate through files with a directory tree.
2.11. A registered user shall be able to organize files alphabetically through a directory tree.
2.12. A registered user shall be able to view the current directory from a *Files* panel.
2.13. A registered user shall be able to create directories.
2.14. A registered user shall be able to create text files.
2.15. A registered user shall be able to create bookmarks of web URLs.
2.16. A registered user shall be able to pin the resources for easy access in the future.
2.17. A registered user shall be able to view supported files on the website.
2.18. A registered user can view the contents of their files on a *Display* panel.
2.19. A registered user can view the contents of their saved web pages on a *Display* panel.

2.21. A registered user shall have a place to type and save a note for each resource.
2.22. A registered user may have an assistant AI chatbot.
2.24. A registered user shall be able to comment on files.
2.25. A registered user shall be able to delete/move resources to a trash bin.
2.26. A registered user shall be able to delete resources from a trash bin permanently.
2.27. A registered user shall be able to recover items from a trash bin.
2.28. A registered user shall be able to download files onto their computer.
2.30. A registered user shall be able to share their directories by configuring the list as ‘public’.
2.31. A registered user shall be able to configure permissions of their shared directories/folders. The permission allows other users to either only view the list or download the list.
2.33. A registered user shall be able to view other directories shared publicly.
2.35. A registered user shall be able to download other shared directories onto their computer.
2.36. A registered user shall be able to search public directories.
2.40. A registered user shall be able to add other registered users as friends.
2.41. A registered user shall be able to remove other registered users from friend-list.
2.42. A registered user shall be able to mark other registered users as their favorite friend.
2.45. A registered user shall be able to configure visibility of their resources at any time. The visibility can be either public or private.
2.46. A registered user shall be able to message another registered user.
2.47. A registered user shall be able to create a group chat to message multiple other registered users.
2.48. A registered user shall be able to search other registered users whose profile is set public.
2.49. A registered user shall be able to search friends.
2.50. A registered user shall be able to search pinned resources.
2.51. A registered user shall be able to clear notes added to the resource.
2.52. A registered user shall be able to clear comments added to the resource.
2.55. A registered user shall be able to rename their resources.

Account

3.1. Account shall have a certain amount of cloud storage.
3.2. Accounts shall require and store a username that is at least 6 characters.

3.3. Accounts shall require and store a real name tied to them, and the real name must have a first and last name.
3.4. Accounts shall require and store a password tied to them.
3.5. Accounts password shall be encrypted when stored in a database.
3.6. Accounts shall be created by one or more general users.
3.7. An account shall be owned by one registered user.

Admin

4.1. Admin shall be able to access all resources, approved or not
4.2. Admin shall be able to delete any resources present on the website
4.5. Admin shall be able to delete/ban any user for any reason (breaking of rules/TOS, inappropriate username)
4.6. Admin shall be able to delete any inappropriate resources uploaded.
4.8. Admin shall check if comments on resources are appropriate.
4.9. Admin shall delete zero or many inappropriate comments on one or many resources.

Resources

5.1. Resources shall be uploaded by a registered user.
5.2. Resources shall be organized in an alphabetical order.
5.3. Resources such as text files shall be able to be created by a registered user.
5.4. Resources shall be viewed by a registered user.
5.5. Resources shall be commented by registered users.
5.6. Resources shall be moved by a registered user to a trash bin.
5.7. Resources shall be able to be deleted permanently by a registered user from a trash bin.
5.8. Resources shall be recovered by a registered user from a trash bin.
5.9. Resources shall be downloaded on the computer by a registered user.
5.10. Resources shall have comments from other registered users.
5.11. Resources shall be able to be accessed by admin.
5.12. Resources shall be able to be deleted by admin.
5.13. Resources shall be deleted if deemed inappropriate by the admin.
5.15. Resources shall have comments verified by admin.
5.16. Resources shall be pinned by a registered user.
5.17. A resource shall have a note created by a registered user.
5.18. Resources shall have configured visibility by a registered user.
5.19. A pinned resource shall be searched by registered users.
5.20. Resources shall be able to be renamed by registered users who added them.

Directory and Directory Tree

8.1. Directory trees shall be navigated by a registered user.
8.2. Directory trees shall have files organized in alphabetical order.
8.3. Directories shall be created by a registered user.
8.5. Directories shall be able to be shared by registered users by configuring them to be 'public.'
8.6. Directories shall have configured permissions added by a registered user.
8.8 Directories shared by configuring it to 'public' shall be able to be viewed by a registered user.
8.10. Directories shall be able to be downloaded by other registered users onto their own computer.
8.11. Directories of other users registered configured as public shall be able to be searched by registered users.

Comment

9.1. Comments shall be added on resources by registered users.
9.2. Comments on resources shall be verified by admin.
9.3. Comments shall be deleted by the admin, if deemed inappropriate.

Friend

10.1. A friend shall be added by registered users.
10.2. A friend shall be removed by registered users.
10.3. A friend shall be added as a favorite by registered users.
10.4. A friend shall be searched by registered users.

Note

11.1 A note shall be typed and saved by a registered user for each resource.
11.3. Notes shall be cleared by registered users at any time.

Priority 2

General User

1.8. A general user shall be able to resend an account verification email.

Registered User

2.29. A registered user shall be able to share their directories via generated link.

2.32. A registered user shall be able to view other directories shared via link.
2.34. A registered user shall be able to download other shared directories into their account.
2.37. A registered user shall be banned for uploading one or many inappropriate resources by admin.
2.38. A registered user shall not be able to view unapproved resources.
2.39. A registered user shall not be able to share unapproved resources.
2.53. A registered user shall be able to disable comments in the resource.
2.54. A registered user shall be able to delete a group chat they created.

Admin

4.3. Admin shall be able to change an unapproved resource to an approved resource.
4.4. Admin shall be able to change approved resources to unapproved resources.
4.7. Admin shall be able to view unapproved resources.

Unapproved Resources

6.1. An unapproved resource shall only be viewed by an admin.
6.2. An unapproved resource shall not be viewable by users until approved.
6.3. An unapproved resource shall be able to be converted to an approved resource by admin.
6.4. Unapproved resources shall not be able to be shared until an admin approves it.

Approved Resources

7.1. Approved resources shall be able to be viewed by admins and users.
7.2. Approved resources shall be able to be converted back to unapproved resources by the admin.
7.3. Approved resources shall be able to be shared via a generated link.

Directory and Directory Tree

8.4. Directories shall be able to be shared by registered users to other registered users via generated link.
8.7. Directories shared via generated links shall be able to be viewed by a registered user.
8.9. Directories shall be able to be downloaded by other registered users onto their own account.

Comment

9.5. Comments shall be able to be cleared by a registered user at any time.
9.6. Comments shall be disabled by a registered user at any time.

Priority 3

General User

1.9. A general user shall be provided a user manual covering basic features.

Registered User

2.2. A registered user shall be able to interact with a skippable tutorial sequence.
2.9. A registered user shall be able to cancel an upload that is in progress.
2.20. A registered user shall be able to change the color theme of the website.
2.23. A registered user shall be able to highlight text on pdf files.
2.43. A registered user shall be able to edit their posted comments at any time.
2.44. A registered user shall be able to edit their saved notes at any time.

Resources

5.14. Resources such as pdf files shall be highlighted by registered users.

Comment

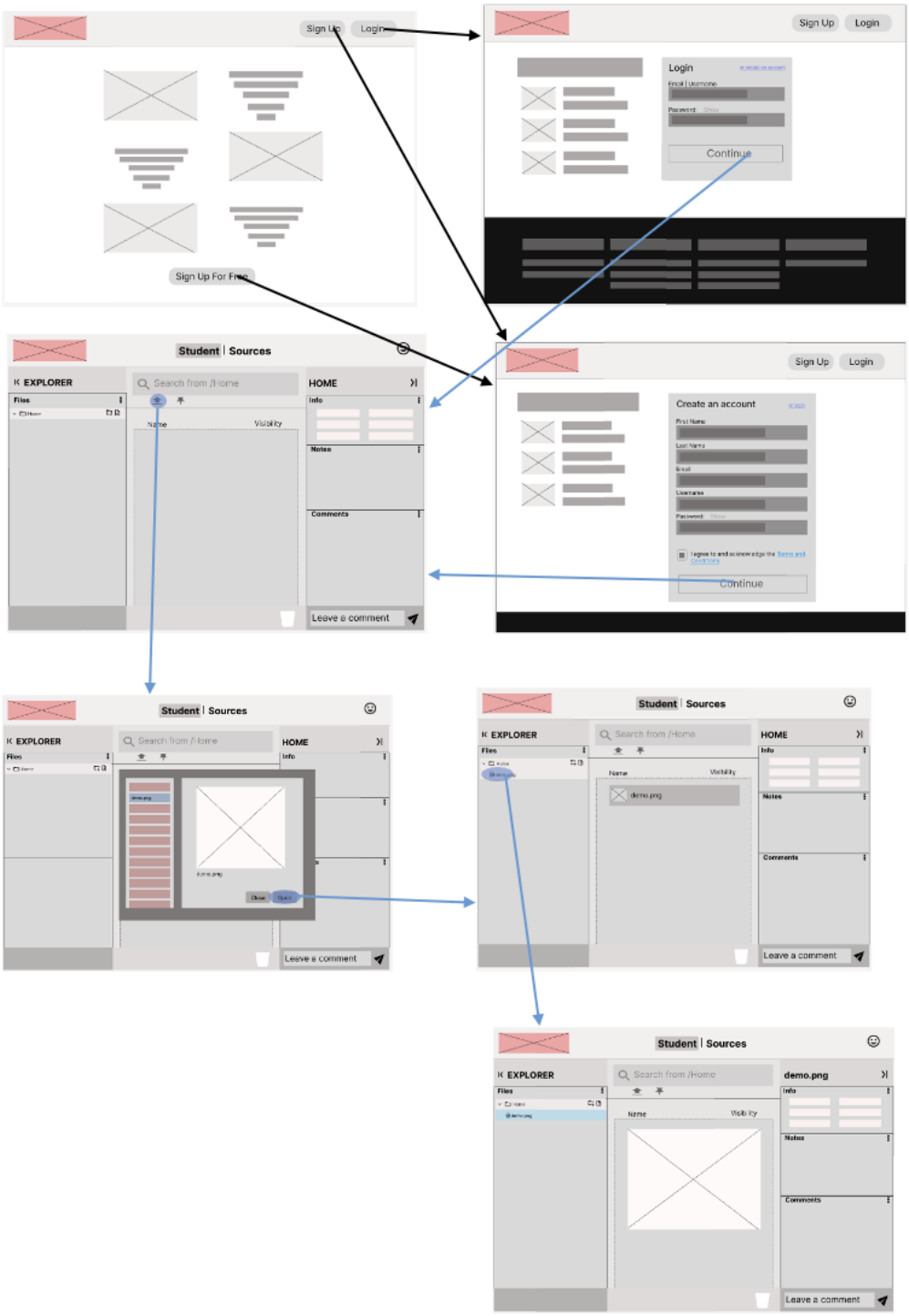
9.4. Comments shall be able to be edited at any time by the registered user who posted it.

Note

11.2. A note shall be able to be edited at any time by the registered user who saved it.

Wireframes

To Upload and Open A File



When a user visits the StudentSources site, they land on our homepage where a short description about the product is given. The user can locate the *Login* button on the top-left corner of the home page or the *Sign Up* button on the top-left corner and bottom of the page, just above the footer. *Login* button directs the user to the login page where the user enters email address or username, along with the password. *The Sign Up* button directs the user to the registration page, where the user enters information such as first and last name, email address, username, and password. The user shall also agree to *Terms and Conditions* to continue the registration process. After successfully logging in or signing in, the user is directed to the *Student* page, where the user can upload, make notes, comment and so on. The page is divided into three sections, left, middle and right sections. The left section has two panels. In the top panel of the left section, the *Files* panel, users can see the directory of the selected folder from where the user can select a file to be displayed/previewed in the middle section. The middle section also works as a platform to view all the folders the user has, configure visibility, upload folders/files. The right section has three panels: *Info, Notes, and Comments.* The *Info* panel displays general information about the selected file/folder. The user can use the Notes panel to add their notes for the selected resource. The user can leave one or many comments on the selected resource with the use of the Comments panel.

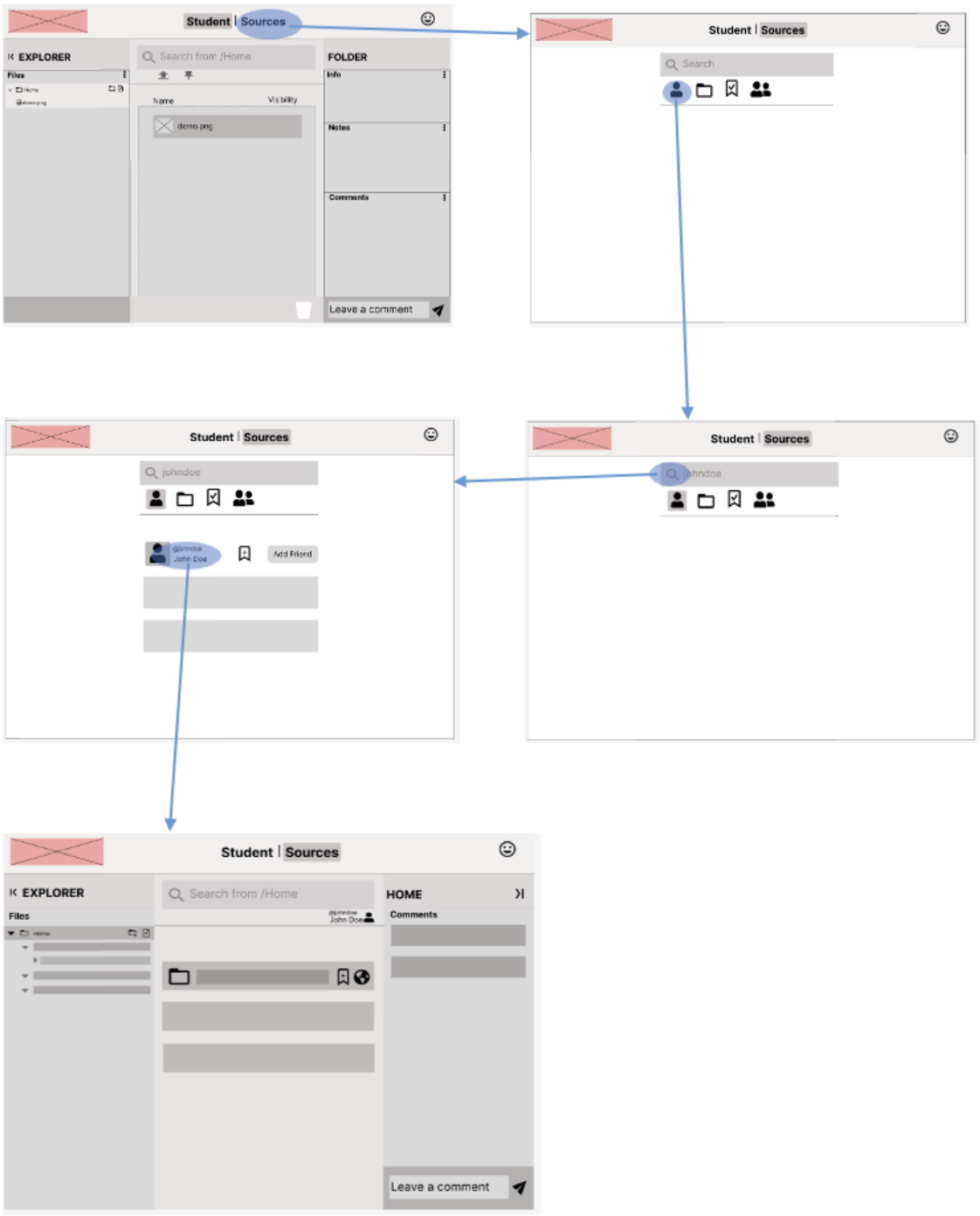
To upload a file, the user can either drag and drop a file or click on the *Upload* button present in the middle Section. Upon clicking the *Upload* button, the user shall be prompted to choose file(s) from their local device and the selected file(s) shall be uploaded to their account. To view the contents of a file, the user can select the file on the *Files* panel, then the file will be opened in the middle section.

To Message Other User



From the previous *Open a File* flow, to exit the file open screen, the user can click on the root directory (*Home*, in this case). The user can find the Inbox feature by navigating to the Nav-Bar and clicking on the *User* icon. After clicking the *Inbox* option, a new window shall be opened where users can create group chats by clicking on the Create *Group Chat* button. The user also has the option to have one-to-one conversation by clicking on the Add icon on the Create *DM* section. To make it easier for users, users can search for other users in order to message them or add to a group chat.

To Search Other User's resources



To search for other users or their resources, the user can click on the *Sources* page located at the top of the current page. The user will be directed to the Search page where they can search with different filter options such as *Users, Folders, Bookmarked or Friends.* Once the search is complete and the results are outputted, the user can select the user they wish to see the resources of.

High Level Database Architecture and Organization

This high level conceptual database design will be based on one of the unique features StudentSources provides, which is the exclusivity to SFSU students and being able to have a dedicated community for SFSU study resources.

Database Organization

Database Requirements:

1. A registered user shall be able to upload zero or many resources.
2. A registered user shall add zero or many comments on their own resources.
3. A registered user shall comment on many other registered users' resources.
4. A registered user shall add zero or many other registered users as friends.
5. A registered user shall remove zero or many other registered users from friend-list.
6. A registered user shall add zero or one note for each resource.
7. A registered user shall set permissions to ‘downloadable’ of many of their shared lists.
8. A registered user shall set permissions to ‘view only’ of many of their shared lists.
9. A registered user shall set visibility as ‘public’ of many of their resources.
10. A registered user shall set visibility as ‘private’ of many of their resources.

Entities:

1. registeredUser
2. userResource
3. resourceComment
4. userFriend
5. sharedList
6. userNote

Entity Attributes and Domains:

1. registeredUser
    a. userID (primary key) = INT, numeric, unique, and not null

    b. username = VARCHAR(32), not null, alphanumeric, can be upper and lowercase, must be unique

c. email = VARCHAR(50), not null, alphanumeric, must contain valid email with a @

d. password = VARCHAR(32), not null, alphanumeric, will be encrypted before storing, must have requirements specified by website

e. formalName = VARCHAR(32), first and last name of user, can only contain letters

f. userImage = BLOB, user profile image saved as binary data

g. sharedListID (foreign key) = INT, numeric, references sharedList's sharedListID

h. resourceID (foreign key) = INT, numeric, references userResources' resourceID

i. commentID (foreign key) = INT, numeric, references resourceComment's commentID

j. noteID (foreign key) = INT, numeric, references userNote's noteID

2. userResource
    a. resourceID (primary key) = INT, numeric, unique, not null, auto incremental

    b. title = VARCHAR(32), not null, alphanumeric, must be unique to prevent confusion of resources.

    c. description = TEXT, can contain anything TEXT data type can store including letters, numbers, special characters. Holds a string with a max length of 65,535 bytes.

    d. uploadDate = DATE, Format: YYYY-MM-DD. The supported range is from '1000-01-01' to '9999-12-31'

    e. courseID = VARCHAR(12), alphanumeric, will be stored as 3-4 letters following 3 numbers, for example: CSC101

f. professorName = VARCHAR(32), alphanumeric, can only contain letters and hyphens

g. isPrivate = TINYINT(1), equivalent to boolean data type, can only contain 0 or 1, tracks if the resource is private or public

h. userID (foreign key) = INT, numeric, references registeredUser's userID

i. commentID (foreign key) = INT, numeric, references resourceComment's commentID

j. noteID (foreign key) = INT, numeric, references userNote's noteID

userResource entity contains child entities which define the format of resources that users store. The child entities with their respective attributes are as follows:

i. textFile
   a. size = INT, numeric
   b. extension = VARCHAR(10), alphanumeric, can contain letters and a special character like '.'

ii. image
   a. size = INT, numeric
   b. extension = VARCHAR(10), alphanumeric, can contain letters and a special character like '.'

iii. video
   a. url = VARCHAR(200), alphanumeric, can contain letters, numbers, and special characters such as ; , / ? : @ & = + $ - _ . ! ~ * ' ( ) #

iv. audio
   a. url = VARCHAR(200), alphanumeric, can contain letters, numbers, and special characters such as ; , / ? : @ & = + $ - _ . ! ~ * ' ( ) #

v. webpage
   a. url = VARCHAR(200), alphanumeric, can contain letters, numbers, and special characters such as ; , / ? : @ & = + $ - _ . ! ~ * ' ( ) #

3. resourceComment
    a. commentID (primary key) = INT, numeric, not null, unique

    b. comment = TINYTEXT, alphanumeric, can contain anything TINYTEXT data type can store including letters, numbers, special characters. Holds a string with a max length of 255 characters.

    c. commentDate = DATE, Format: YYYY-MM-DD. The supported range is from '1000-01-01' to '9999-12-31'

    d. resourceID (foreign key) = INT, numeric, references userResource's resourceID field, creating a relationship between the resource and the comment that was posted on the resource.

    e. userID (foreign key) = INT, numeric, references registeredUser's userID field, creating relationship between the comment and the user who commented it.

4. userFriend
    a. UID1 (primary key and foreign key) = INT, numeric, not null, references registeredUser's userID
    b. UID2 (primary key and foreign key) = INT, numeric, not null, references registeredUser's userID

5. sharedList
    a. sharedListID (primary key) = INT, numeric, not null

    b. url = VARCHAR(200), alphanumeric, can contain letters, numbers, and special characters such as ; , / ? : @ & = + $ - _ . ! ~ * ' ( ) #

    c. isDownloadable = TINYINT(1), equivalent to boolean data type, can only contain 0 or 1, tracks if the resource is view only or is downloadable by other registered users

    d. userID (foreign key) = INT, numeric, references registeredUser's userID

    e. resourceID (foreign key) = INT, numeric, references userResource's resourceID

6. userNote
    a. noteID (primary key) = INT, numeric, not null,

    b. note = TINYTEXT, alphanumeric, can contain anything TINYTEXT data type can store including letters, numbers, special characters. Holds a string with a max length of 255 characters.

    c. userID (foreign key) = INT, numeric, references registeredUser's userID

    d. resourceID (foreign key) = INT, numeric, references userResources' resourceID

Entity Relationships:

1. One-to-Many (1:M), registered user to resources
    a. A resource shall be owned/posted by **only one** user.
    b. A user shall be able to post **many** resources.

2. One-to-Many (1:M), registered user to resource comments
    a. A comment shall be posted by **only one** user.
    b. A user shall be able to post **many** comments on a resource.

3. One-to-Many (1:M) registered user to user friend
    a. A friend shall be associated with **only one** user. For example, when user A sends a friend request to user B and the request is accepted, the connection shall be only between those two users.
    b. A user shall be able to accept **many** friend requests and become friends.

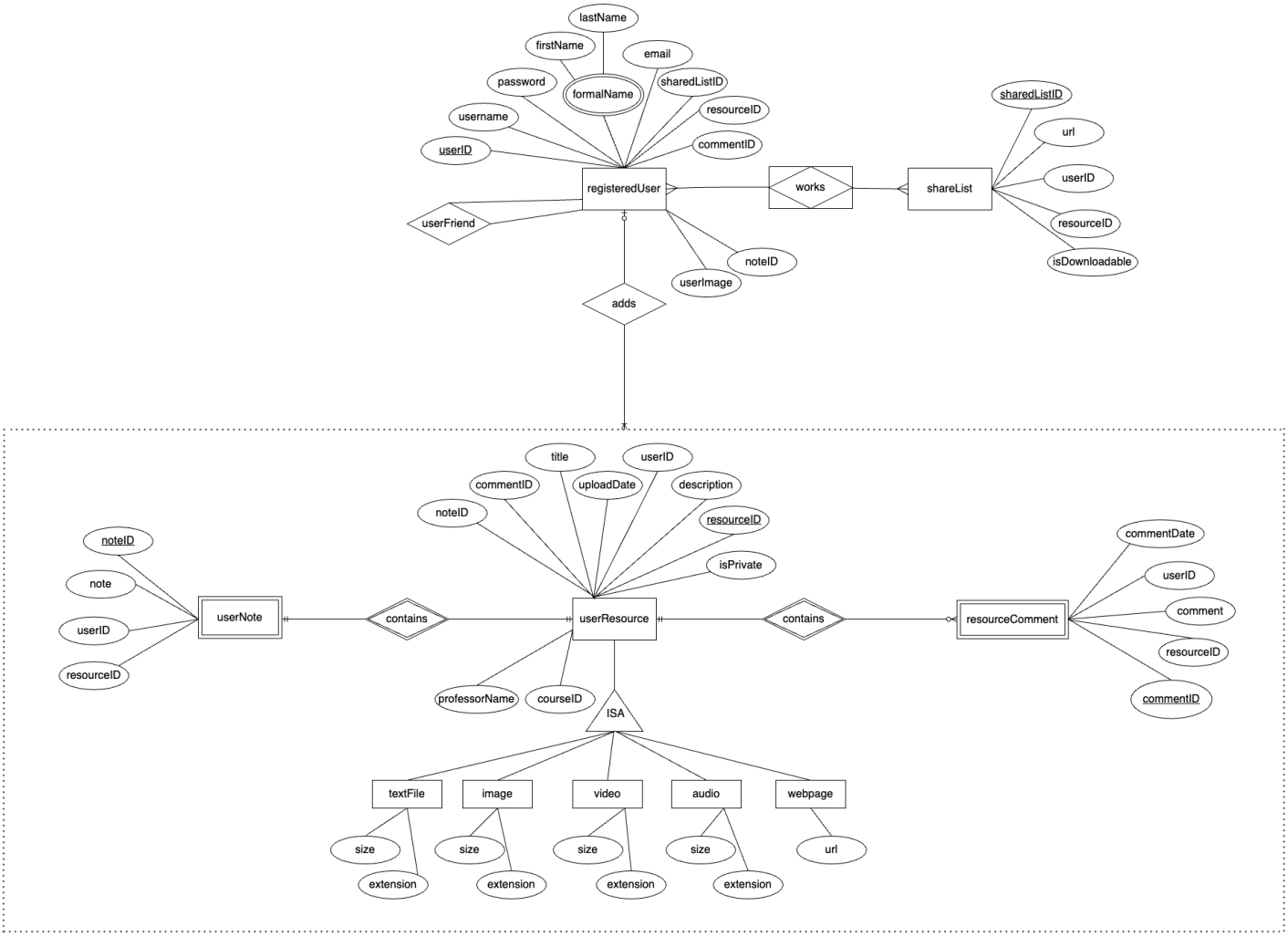
4. Many-to-Many (M:M) registered user to shared list
    a. A registered user shall work on **many** shared lists.
    b. A shared list shall be worked on by **many** registered users.

5. One-to-One (1:1) registered user to note
    a. A note shall be associated with **only one** resource.
    b. A resource shall be able to contain **only one** note.

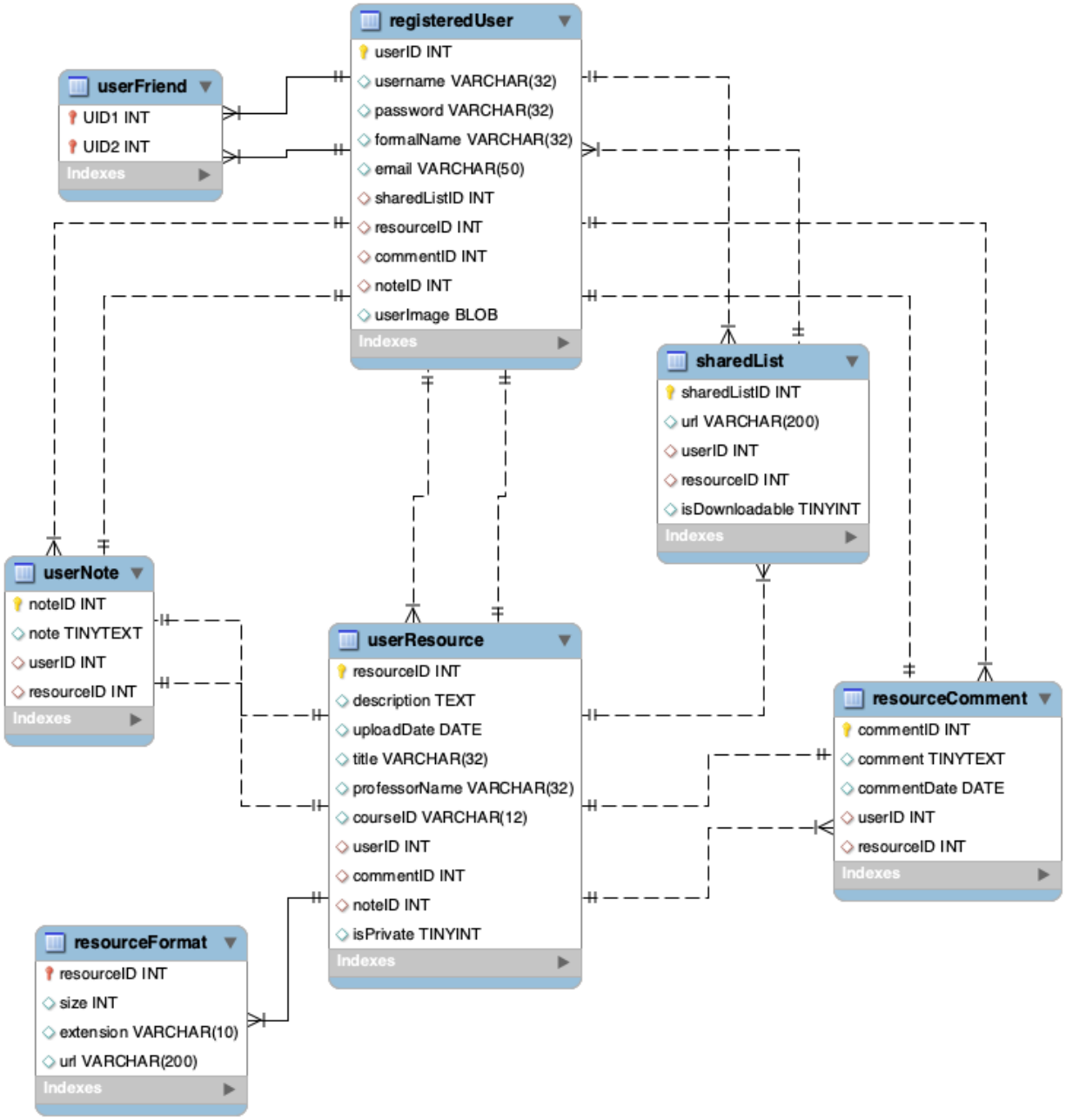
6. One-to-Many (1:M) resource to resource comments
    a. A comment shall be associated with **only one** resource.
    b. A resource shall be able to contain **many** comments.

7. One-to-One (1:1) resource to user note
    a. A resource shall have **only one** user note.
    b. A user note shall be added to **only one** resource.

Entity Relationship Diagram (ERD)



Enhanced Entity Relationship (EER) Diagram



Database Management System (DBMS)

The DBMS we chose for database management is MySQL because we are familiar with its syntax; in addition, it is simple and efficient.

Media Storage

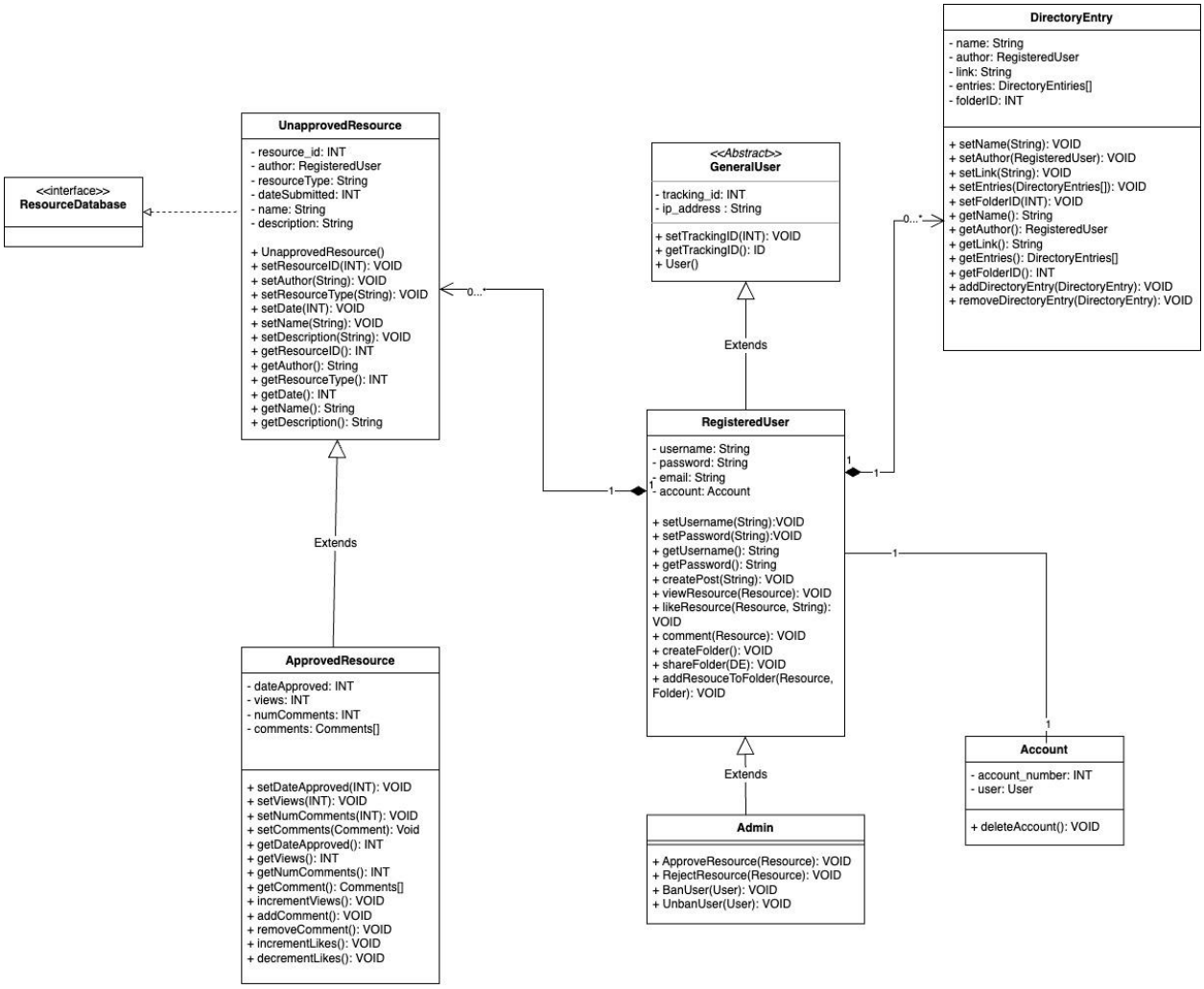
The multimedia contents will be stored in a mountable file system. The data format for images are jpeg and png, for audio files are mp3 and for video files are mp4.

Search/filter architecture and implementation

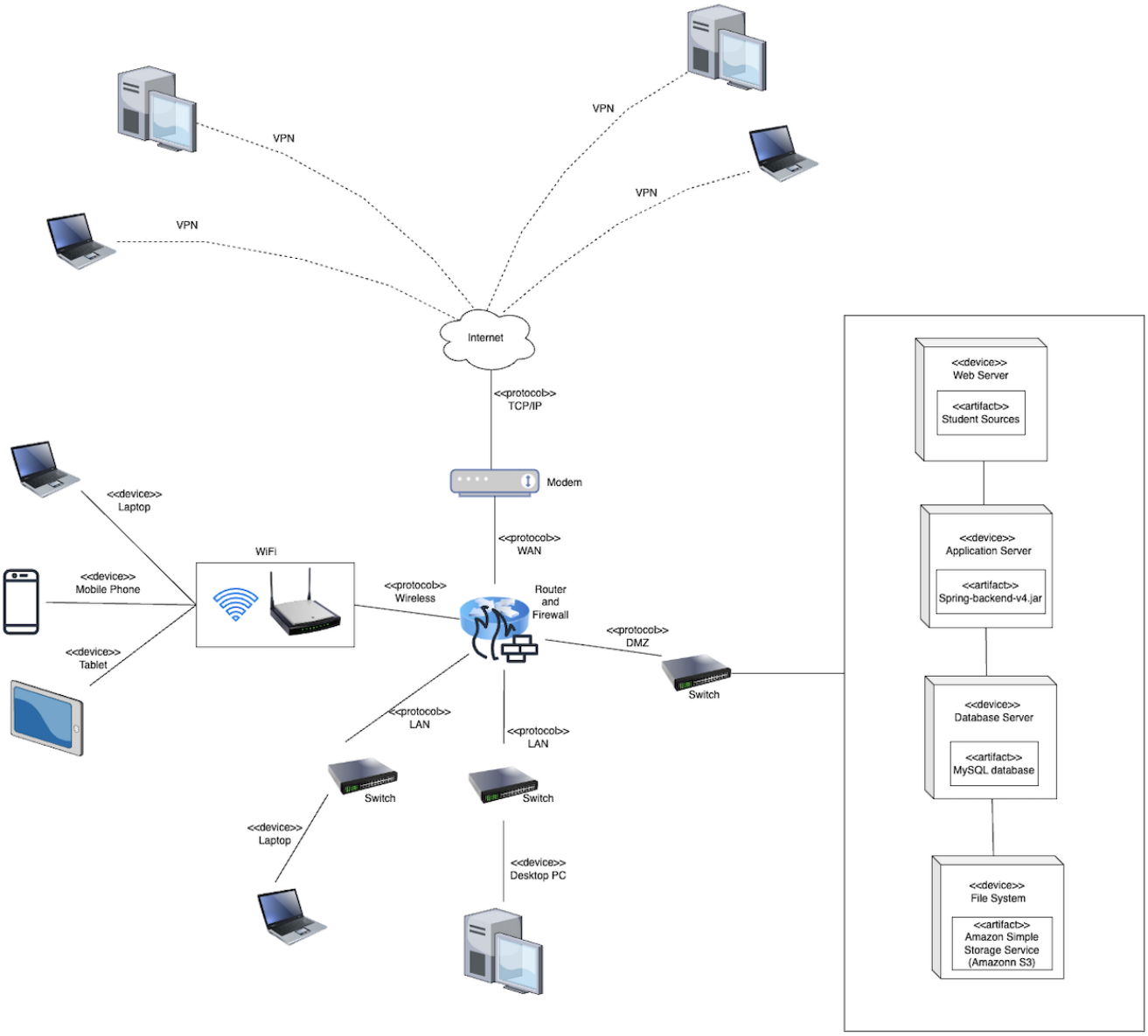
The algorithm used to search for registered users in the database starts in the user repository in the searchUsersByUsername method with the SQL statement “SELECT * FROM user WHERE user LIKE :query OR first_name LIKE :query OR last_name LIKE :query”. This statement, which will select entities with identical values in the user, first_name, or last_name attributes, is recognized as a query due to Spring Boot’s Query annotation and then stored into a string parameter called query. The Query is then modified to handle white spaces and then stored into another string variable called newQuery, which is then passed into searchUsersByUsername and returned as an User entry. After that, both the password and email of said user are encrypted via for loop and the entry is then returned if it exists and will return null if it is not.

High Level Diagrams V2

UML Diagram



Deployment and Network Diagram



List of Team Member Contributions

Team Member	**Contribution**
Saru Shrestha	- Documentation - Wireframe - Update ERD - EER diagram - Update UML - Update Networks and Deployment Diagram - Gather image contents - Reference page (frontend) - Revise M3 documentation
Kilian Kistenbroker	- Plan and assign tasks for frontend - Render directory tree in left panel - Render current directory onto main panel - Implement functionality of ‘Create Folder’ - Implement functionality for uploading through drag-and-drop and upload button - Render file viewer of different file format - Implement functionality for downloading files from file viewer into users local downloads folder - Implement functionality for ‘pinning’ folders and files - Implement functionality for ‘create file’ for txt and url files - Implement search functionality for folders/files in a current directory - ‘Info’ tab that displays basic information of current folder/file - Render other users homepage (using dummy data) - General purpose pop-up windows for displaying warnings or errors that may occur - Review and debug code - Update names in database to be consistent with documentation - Integrate frontend, backend and

	database
Elias Abay	- Plan and assign backend task - Research file system - Configure file system - Generate root folder for newly registered users - Implement functionality to delete items from root folders - Review code - Integrate frontend, backend and database
Parth Panchal	- ‘Notes’ tab (frontend) - ‘Comments’ tab (frontend) - Implement functionality for ‘notes’ tab (backend) - Implement functionality for ‘comments’ tab (backend) - Maintain backup GitHub repository
Joshua Del Rosario	- Research file system - Configure file system - Implement functionality to upload and download files and folders in the file system - Implement functionality to return root folder of other registered users - Update names in database to be consistent with documentation - Revise ERD diagram
Wilson Yin	- Research file system - Configure file system - Implement functionality to retrieve files and folders from file system to the user end
Omar Thiongane	- ‘Notifications’ Panel (frontend) - ‘User Account Panel’ with features of logging out, displaying remaining storage, and displaying user info (frontend) - ‘Visibility and permissions’ Panel - Take notes during in-person review of prototype

Team Member Contribution Score

Team Member	**Score**
Kilian Kistenbroker	10
Elias Abay	9
Parth Panchal	9
Joshua Del Rosario	9
Wilson Yin	9
Omar Thiongane	9

Scale: 1 to 10, where a score of 1 means “no work at all”, and score of 10 means “contributed to the maximum capacity”